



Predictive Modeling of Agricultural Commodity Prices in Davao City Using Weather and Market Data

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Table of Contents

1	Introduction	2
1.1	Background of the Study	2
1.2	Problem Statement	3
1.3	Objectives of The Study	4
1.4	Significance of The Study	4
1.5	Scope and Limitations of the Study	5
2	Review of Related Literature	7
2.1	Definition of Variables and Problem Formulation	7
2.1.1	Input Feature Space (X) and Output Target Variable (Y)	7
2.1.2	The Predictive Mapping Function ($f : X \rightarrow Y$)	8
2.1.3	Formal Definition of the Shock Regime	8
2.2	Agricultural Commodity Price Volatility and Climate Anomalies	9
2.2.1	Theoretical Foundations of Price Volatility	9
2.3	Limitations and Conventional Approaches	10
2.4	The Hybrid Regime-Switching Approach	11
2.4.1	Principles of Learning-Based Hybrid Models	11
2.4.2	Efficacy of Gradient-Boosted and Deep Learning Ensembles	12
2.4.3	Adaptation to the Davao City Agricultural Portfolio	12
2.4.4	Computational Mechanics of the Regime-Switching Model	13
2.4.5	Temporal Disaggregation	13
2.4.6	Heuristic Feature Optimization	14
2.4.7	Extreme Gradient Boosting (XGBoost) for Shock Modeling	15
2.4.8	Regime Identification and Ensemble Weighting	16
2.5	Summary and Research Gap	17
2.5.1	Summary of Key Research Studies	17
2.5.2	Research Gap and Justification	18
2.6	Theoretical Framework	19
2.6.1	Theory of Temporal Disaggregation	19
2.6.2	Evolutionary Theory of Heuristic Feature Optimization	20
2.6.3	Theory of Regime-Switching Dynamics and Market States	21
2.6.4	Theoretical Mechanics of Dual-Regime Hybrid Ensembles	22
2.6.5	Theoretical Basis for Dual-Metric Evaluation	23
3	Methodology	25

3.1 Conceptual Framework 25

3.1.1 Phase 1: Data Resolution 25

3.1.2 Phase 2: Dimensionality Reduction 25

3.1.3 Phase 3: Regime-Switching Logic 27

3.1.4 Phase 4: Output and Validation 27

3.2 Research Design 28

3.3 Data Collection 28

List of Figures

1	Theory of Temporal Disaggregation	19
2	Evolutionary Theory of Heuristic Feature Optimization	20
3	Theory of Regime-Switching Dynamics and Market States	21
4	Theoretical Mechanics of Dual-Regime Hybrid Ensembles	22
5	Theoretical Basis for Dual-Metric Evaluation	23
6	Conceptual Framework of the Dual-Regime Predictive Engine	26

List of Tables

1	Summary of Methodologies and Key Findings	17
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Abstract

Chapter 1

Introduction

1.1 Background of the Study

Agricultural commodity prices in Davao City are highly unpredictable. The prices of vegetables and other farm products can change dramatically in a short period, making it difficult for farmers, traders, and policymakers to plan. Unlike stable markets, the Davao Region is frequently exposed to extreme weather events such as typhoons, which can destroy crops and block the roads used to transport goods to the market [16, 17]. When supply chains are disrupted in this way, the prices of basic food items can spike sharply in a matter of days. For instance, consecutive typhoons in 2024 and 2025 caused billions of pesos in agricultural damage across Mindanao, triggering vegetable price surges of up to 122 percent in affected areas [17]. The problem is that these sudden price changes are very difficult to predict with the tools currently available [12, 15].

One reason why predicting these price changes is so difficult is the way agricultural data is collected and reported. Government records on commodity prices are usually published as monthly averages. When a typhoon causes a price spike that lasts only one or two weeks, that spike gets hidden in the monthly average and is no longer visible in the data. Moauro and Savio [11] treats this as a "missing observation problem", and Moauro and Savio [11] further establish that when high-frequency shocks are absorbed into low-frequency aggregates, the statistical integrity of the original disruption is lost. Compounding this data problem is the structural weakness of the tools most commonly used to forecast prices. Traditional approaches such as ARIMA and SARIMA assume that prices change in a smooth and predictable pattern [21, 22]. As Sun F. et al. [21] and Mulla and Usha [12] have consistently shown, these linear models are simply not equipped to handle the sudden, nonlinear price spikes that follow typhoon-induced supply chain disruptions. Because of this, the forecasting challenge in Davao City is twofold: the data does not capture the full impact of short shocks, and the tools used to read that data are not designed for the kind of disruptions the region regularly experiences.

Several researchers have tried to address these limitations by developing more advanced forecasting models. To recover the high-frequency price signals that are hidden inside monthly data, Moauro and Savio [11] established a multivariate temporal disaggre-

gation framework that uses related high-frequency indicator variables to estimate what monthly figures actually looked like on a week-by-week basis, a method later extended by Mosley et al. [7] for high-dimensional datasets. As part of this process, both daily weather data from PAGASA and macroeconomic indicators, specifically weekly fuel prices and the consumer price indices, are prepared as matching high-frequency indicator variables alongside the disaggregated weekly price data [11, 17]. Fuel prices are included because rising transport costs directly increase the retail price of crops moved along Davao’s road networks, while the consumer price indices capture the broader inflationary environment that shapes the baseline level of food prices independent of weather events. To handle the nonlinearity problem, Zhao et al. [26] demonstrated that combining a Temporal Convolutional Network with XGBoost produced significantly more accurate forecasts for agricultural market data compared to standalone ARIMA models, with the XGBoost component proving especially effective at capturing non-linear price threshold effects. Similarly, Pandit et al. [14] validated a hybrid model that reduced forecasting errors by 20–62% relative to ARIMA benchmarks across several commodity types. Additionally, Chatterjee et al. [4] and Choudhary et al. [3] demonstrated that Genetic Algorithms can effectively identify the most useful input variables for forecasting, reducing noise and improving model reliability across varying seasonal and climatic conditions.

These approaches offer significant advantages when combined into a single hybrid framework. Patro et al. [15] found that integrating a linear baseline model with XGBoost within a multi-factor framework yields a robust ensemble capable of handling both stable and severely disrupted market conditions. The regime-switching dimension of this architecture, supported by the work of Montaña et al. [5], who showed that Markov-based switching models can accurately detect structural volatility transitions triggered by Philippine climate events such as El Niño, La Niña, and typhoons, provides a dynamic gating mechanism that activates the non-linear XGBoost learner precisely when a climate threshold is breached. By first applying temporal disaggregation to recover weekly price signals from monthly records [11, 7], then using a Genetic Algorithm to select the most relevant weather and market features [3, 4], and finally deploying this climate-gated dual-regime engine, the resulting architecture can produce forecasts that are both accurate during calm periods and reliable during climate disruptions [1, 14, 26]. This makes it particularly well-suited for a region like Davao City, where typhoon-driven supply shocks are a regular and damaging occurrence [16, 17].

1.2 Problem Statement

This study seeks to investigate the problem of predicting sudden, typhoon-driven agricultural commodity price shocks in Davao City, where existing forecasting tools are unable to detect the non-linear price spikes that follow climate-induced supply chain disruptions [11, 22]. A central challenge is the “missing observation problem” inherent in official monthly price records. When a typhoon-driven price spike lasts only one or two weeks, its statistical impact is absorbed into the monthly aggregate and effectively disappears from the data [11]. This structural data gap, combined with the inability of linear models to capture non-linear threshold events, means that the true economic impact of short-duration supply chain shocks remains systematically obscured.

Specifically, the study sought to answer the following questions:

1. What is the impact of extreme climate anomalies, specifically typhoons, on the localized price volatility of the Davao City vegetable portfolio?
2. How does the reliance on low-frequency monthly agricultural price aggregates obscure the true economic impact of short-duration supply chain shocks?
3. To what extent can multivariate temporal disaggregation reconstruct a high-resolution weekly price series from official monthly records using high-frequency indicator variables?
4. Which climate-price lag combinations, identified through Genetic Algorithm-based feature optimization, yield the strongest predictive signal for typhoon-driven price movements?
5. How accurately does a dynamic regime-switching ensemble, utilizing a SARIMAX baseline and an XGBoost non-linear learner, forecast sudden price jumps triggered by exogenous climate thresholds compared to standalone benchmark models?

1.3 Objectives of The Study

The primary objective of this study is to develop a dynamic, regime-switching predictive architecture capable of accurately forecasting agricultural commodity prices in Davao City by integrating classical econometrics with non-linear machine learning, governed by real-time climate thresholds.

1. To quantify the historical impact of extreme climate anomalies, specifically typhoons, on the localized price volatility of the Davao City vegetable portfolio.
2. To mathematically resolve the "missing observation problem" inherent in low-frequency monthly agricultural records, unmasking short-duration supply chain shocks.
3. To implement multivariate temporal disaggregation, utilizing PAGASA daily meteorological data, weekly fuel prices, and consumer price index as high-frequency indicator variables to reconstruct a high-resolution weekly price series.
4. To deploy a Genetic Algorithm (GA) for heuristic feature optimization, isolating the most predictive climate-price lag combinations while discarding spectral data noise.
5. To develop and evaluate a dynamic regime-switching ensemble, utilizing a SARIMAX baseline for routine seasonality and an XGBoost learner for threshold-driven shocks, rigorously validated using Root Mean Squared Error (RMSE) and Mean Absolute Percentage Error (MAPE).

1.4 Significance of The Study

Accurate forecasting of agricultural commodity prices is essential for the economic stability of farming communities, especially in regions frequently affected by typhoons and

other extreme weather events [16, 17]. In Davao City, sudden price spikes can leave farmers and consumers without enough time to prepare, causing significant financial and food security problems. By developing a hybrid forecasting model that can detect and respond to climate-driven price shocks, this study provides a practical tool for addressing one of the most pressing challenges in local agricultural planning.

For farmers and agricultural cooperatives, the forecasting system produced by this study can serve as an early-warning tool. By anticipating price movements before they occur, farmers can make better decisions about when to harvest, how to store their produce, and how to protect their crops from incoming weather events [15, 18]. This kind of advance information can reduce losses and help stabilize incomes during periods of climate disruption.

For logistics planners and supply chain operators, the model provides data-driven insights to reroute distribution networks and manage buffer stocks before typhoon-induced bottlenecks trigger retail price crises [15, 16]. Knowing in advance that a price shock is likely to occur gives operators the time needed to make adjustments and maintain market stability.

1.5 Scope and Limitations of the Study

The study covers the predictive modeling of monthly agricultural commodity prices in Davao City, Philippines, with a focus on vegetables and other climate-sensitive crops traded in the local market. The data used includes official agricultural price records from the Philippine Statistics Authority and daily meteorological data from the Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAGASA) for the Davao Region [13], weekly fuel price data, and consumer price index published by the Philippine Statistics Authority (PSA). These macroeconomic indicators serve as high-frequency exogenous variables within the temporal disaggregation stage, enabling the model to account for supply chain cost pressures and broader inflationary trends that influence retail commodity prices independently of weather events. The proposed modeling framework integrates multivariate temporal disaggregation [11, 7], Genetic Algorithm-based feature selection [3, 4], a SARIMAX linear baseline, and an XGBoost non-linear learner [14, 26] governed by a climate-triggered gating mechanism [5]. Model performance is evaluated using Root Mean Squared Error (RMSE) and Mean Absolute Percentage Error (MAPE) [9, 26] across the full commodity portfolio.

The study is subject to several limitations. First, the quality of the temporal disaggregation output depends on the completeness and consistency of the available daily weather and economic indicator data; any gaps in PAGASA records or fuel price data may affect the accuracy of the reconstructed weekly price series [11, 7]. Second, the climate thresholds used to trigger the regime-switching mechanism are based on historical data, which means the model may not perform equally well for weather events that fall outside the training period [21]. Third, the model is specifically calibrated for the Davao City market and may require adjustments before it can be applied to other regions with different agricultural and climatic conditions [12]. Finally, the study does not account for sudden policy changes or international commodity market shocks that are unrelated

to local weather patterns, as these are outside the scope of the weather-and-market-data framework used in this research [12, 21].

To ground the proposed architecture in established theory and empirical evidence, the following chapter presents a systematic review of the related literature. The review is organized thematically, beginning with the theoretical foundations of climate-driven price volatility, evaluating the limitations of conventional forecasting approaches, and synthesizing the computational methodologies, temporal disaggregation, Genetic Algorithm optimization, and the Hybrid Regime-Switching Ensemble that collectively address the research gaps identified in this chapter

Chapter 2

Review of Related Literature

2.1 Definition of Variables and Problem Formulation

To create a clear and precise foundation for the theoretical and computational arguments presented in this chapter, this section formally defines the variables, feature spaces, and mathematical constructs used throughout the study. The notation introduced here is applied consistently across both Chapter 2 and Chapter 3.

2.1.1 Input Feature Space (X) and Output Target Variable (Y)

The input feature space X is a multivariate matrix of high-frequency weekly observations drawn from three categories of exogenous and endogenous indicators. Formally:

$$X = X_{climate}, X_{macro}, X_{price}$$

where:

- $X_{climate}$ represents the set of weekly climate features derived from daily PAGASA meteorological observations for the Davao Region, including total weekly rainfall (mm), maximum sustained wind speed (km/h), and mean weekly temperature (°C), temporally aggregated from daily records.
- X_{macro} represents the set of weekly macroeconomic control variables, specifically weekly retail fuel prices (PHP/liter) and the Consumer Price Index (CPI), sourced from the PSA and DOE, which capture supply chain cost pressures and the broader inflationary environment independently of weather events.
- X_{price} represents the set of lagged commodity price features, specifically the disaggregated weekly retail prices of the Davao City vegetable portfolio reconstructed from official PSA monthly records through multivariate temporal disaggregation.

The Genetic Algorithm optimization layer described in Section 2.4.6 selects the most predictive subset of features from X , producing an optimized feature vector X^* that is passed to the predictive engine.

The output target variable Y is defined as the predicted weekly retail price for a specific agricultural commodity in Davao City, measured in Philippine Pesos per kilogram (PHP/kg). Formally:

$$Y_t = f(X_t)$$

where Y_t is the forecast price at week t , and X_t is the corresponding optimized feature vector at the same time step.

2.1.2 The Predictive Mapping Function ($f : X \rightarrow Y$)

The central objective of this study is to learn a mapping function:

$$f : X \rightarrow Y$$

The goal is for f to accurately approximate the actual weekly commodity price during both stable periods and market shocks caused by typhoons. Instead of using a single static function, f is designed as a dynamic dual-regime ensemble controlled by a climate-triggered gating mechanism, which is defined as:

$$f(X_t) = \partial_1 \cdot g_{\text{stat}}(X_t) + \partial_2 \cdot g_{\text{ml}}(X_t)$$

where g_{stat} is the SARIMAX linear baseline component, g_{ml} is the XGBoost non-linear learner component, and ∂_1 and ∂_2 are modulating ensemble weights whose values are dynamically adjusted by the Regime Identification Gate described in Section 2.4.8.

2.1.3 Formal Definition of the Shock Regime

The regime-switching mechanism is controlled by a binary indicator function, I_{shock} , which divides the market into two separate operating states. In this framework, P_t represents the observed weekly precipitation at time t , and W_t represents the maximum sustained wind speed at time t , with both datasets obtained from regional PAGASA stations. The shock indicator is defined as:

$$I_{\text{shock}}(t) = 1 \text{ if } P_t \geq \theta_P \text{ or } W_t \geq \theta_W, \text{ else } 0$$

In this equation, θ_P represents the critical weekly precipitation threshold and θ_W is the critical wind speed threshold. Both values are calibrated using historical PAGASA records for the Davao Region to match conditions associated with typhoon-level events.

When $I_{\text{shock}}(t) = 0$, the system operates in the normal regime ($S_t = 0$), where the ensemble weighting prioritizes the SARIMAX baseline such that $\partial_1 \gg \partial_2$.

When $I_{\text{shock}}(t) = 1$, the system transitions into the shock regime ($S_t = 1$), and the ensemble weighting shifts to favor the XGBoost non-linear learner, meaning $\partial_2 \gg \partial_1$.

This formal definition operationalizes the conceptual regime-switching framework established by Montaña et al. [5] and Yuan and Kwon [24] within the specific climatic and agricultural context of Davao City.

2.2 Agricultural Commodity Price Volatility and Climate Anomalies

This chapter provides a systematic review of the literature on climate-driven agricultural price volatility and the modeling frameworks used to predict these non-linear market shifts. To establish a solid theoretical foundation for the study, the review is organized thematically.

First, it examines the broader issue of agricultural commodity price volatility, specifically focusing on how climate anomalies and infrastructure disruptions impact supply chains. Second, the chapter evaluates the effectiveness of current forecasting methods, ranging from traditional econometric models to machine learning approaches, while highlighting their limitations in shock-prone markets. Finally, the review synthesizes these methodologies within a hybrid regime-switching framework, establishing the theoretical justification for using Extreme Gradient Boosting (XGBoost) and temporal disaggregation to capture sudden, threshold-driven price shocks in Davao City's agricultural market.

2.2.1 Theoretical Foundations of Price Volatility

In modern econometric and agricultural research, commodity price volatility is widely viewed as the difficulty of connecting external environmental factors to non-linear economic outcomes. This complexity has historically restricted frameworks that relied solely on linear relationships [9, 22]. In the past, researchers often approached this issue through a linear lens, explaining price fluctuations primarily through seasonal cycles, historical trends, and autocorrelation [22]. More recently, macro-level frameworks have redefined the problem by adopting non-linear approaches, which are better suited to capturing the complex and multifaceted impact of environmental changes on economic results [6].

As a result, the primary issue identified in current research is that traditional linear models are structurally unable to capture the severe market disruptions caused by extreme weather events [22, 13]. As established by researchers such as Sun et al. [22] and Mulla and Usha [12], non-linear physical disruptions, including sudden climate shocks, act as external factors that fundamentally change the statistical behavior and stability of price data. Therefore, the forecasting challenge is no longer simply about predicting a continuous price trend. Instead, it requires accurately modeling specific "threshold" events where minor weather changes allow for market stability, but extreme climatic anomalies trigger disproportionately large spikes in agricultural prices [15].

While global models focus on broad seasonal trends, agricultural price volatility in the Philippines, specifically in the Davao Region, is primarily driven by "extreme shocks" caused by frequent climate anomalies [18, 19]. Climate assessments from PAGASA explicitly identify the Davao Region as a highly vulnerable zone for these types of agricultural disruptions [13]. Recent extreme weather events illustrate this non-linear impact; for instance, consecutive typhoons in 2024 and 2025 caused billions of pesos in damage to infrastructure and crops, disrupting farm-to-market logistics and triggering vegetable price hikes of up to 122% across Mindanao [17, 18, 7, 8]. Consequently, retail prices in the region often show a "lagged" spike, as logistical bottlenecks intensify the effects of the

initial physical crop damage [17, 19].

Adding to this physical vulnerability is a significant technical hurdle known in econometric literature as the "missing observation problem" [7]. Standard government reporting systems often obscure frequent, typhoon-driven price spikes by averaging them into monthly figures [7]. For instance, if a severe supply shock lasts for only one week, its statistical impact is largely lost within a thirty-day aggregate [11]. This lack of detail creates blind spot for data, making it difficult for standard forecasting models to identify the true, high-resolution link between local climate anomalies and the resulting market volatility in Davao's vegetable portfolio [21, 18].

2.3 Limitations and Conventional Approaches

Historically, agricultural price forecasting has relied on classical econometric models, most notably the AutoRegressive Integrated Moving Average (ARIMA) and its seasonal extension, Seasonal ARIMA (SARIMA) [20]. ARIMA operates by forecasting future values through a linear combination of past observations (autoregression) and past forecast errors (moving average) after differencing the data to achieve stationarity [14], while SARIMA expands this architecture to explicitly capture recurring annual harvest cycles. Because these frameworks are fundamentally built on the assumption that future price movements can be linearly explained by historical trends, seasonal patterns, and autocorrelation, they provide a solid baseline for predicting routine and stable seasonality [12, 21]. However, they possess a major drawback: their strict reliance on stationarity and linearity. As established by Sun et al. [22] and Mulla and Usha [13], classical time-series models consistently fall short when dealing with sudden structural breaks. In highly volatile and climate-sensitive markets like the Davao Region, the non-linear price surges triggered by typhoon-induced infrastructure failures are simply beyond the modeling capacity of these traditional ARIMA and SARIMA approaches [17, 12].

To address these linear limitations, researchers have employed time-series decomposition techniques, such as Seasonal-Trend decomposition using Loess (STL), to separate predictable seasonal cycles from irregular, random movements [23, 25]. However, relying solely on decomposition introduces a significant methodological flaw known as the "smoothing effect." As Zhang et al. [26], observe, aggressive smoothing can accidentally merge short-term, extreme price spikes into the general trend. Consequently, instead of accurately isolating sudden anomalies, this process can lead to over-decomposition and an excessive number of residual components, which may increase errors when forecasting complex, non-linear market shifts [25]. Since neither strictly linear models nor basic decomposition methods can sufficiently isolate and predict threshold-driven market disruptions, a shift toward more adaptive, non-linear modeling is necessary [25, 26].

In order to address the limitations of linear econometric models and the risk of losing data signals in traditional decomposition, recent research has increasingly turned toward Learning-Based Hybrid Approaches [1]. Specifically, a Regime-Switching Ensemble Architecture provides the flexibility required to model the complex, dual nature of agricultural price volatility [5]. By combining a linear baseline, specifically SARIMAX, which

extends the seasonal ARIMA structure to incorporate exogenous climate and macroeconomic regressors, to track regular seasonal trends alongside an Extreme Gradient Boosting (XGBoost) component to handle non-linear shocks, this architecture can adapt dynamically to shifting market conditions [15].

Unlike static frameworks, a regime-switching model uses a gating mechanism triggered by high-frequency climate indicators [5, 15]. This ensures that threshold events, such as supply chain disruptions caused by typhoons, activate the non-linear shock model, allowing it to capture sudden price spikes rather than smoothing them away. The following sections evaluate the literature surrounding this hybrid framework, establishing the theoretical and methodological foundation for its application to the Davao City vegetable portfolio.

2.4 The Hybrid Regime-Switching Approach

Having established the structural weaknesses of traditional linear forecasting during extreme climate shocks, this section provides a thorough evaluation of the Learning-Based Hybrid Approach. It focuses specifically on the Regime-Switching Ensemble Architecture, which serves as the primary theoretical framework for addressing the dual-natured volatility analyzed in this research [5, 15].

Drawing on current literature, this section outlines the fundamental principles of hybrid modeling and reviews key studies that have applied these methods to volatile agricultural markets. It also evaluates the specific techniques used in those works, demonstrating their direct relevance and applicability to the local challenges facing Davao City’s vegetable portfolio.

2.4.1 Principles of Learning-Based Hybrid Models

Learning-based hybrid models are built on the understanding that agricultural price volatility has a dual nature, consisting of both predictable seasonal patterns and unpredictable, threshold-driven shocks [26, 9]. Current literature suggests that no single algorithmic framework can effectively manage both stable and volatile market conditions simultaneously [26]. While traditional linear models are well-suited for capturing routine trends and seasonality, they are unable to process sudden structural market disruptions [5]. In contrast, advanced non-linear systems—such as gradient-boosted trees or deep learning networks—are highly effective at modeling complex threshold effects, yet they may overfit to noise during calm, non-volatile periods [26, 3].

To bridge this gap, learning-based hybrid approaches combine these distinct models into a unified ensemble. Rather than treating seasonality and shocks as unrelated issues, hybrid architectures use a transparent linear model to set a baseline forecast, while a machine learning component is specifically used to learn the non-linear residual errors [15]. Additionally, when equipped with a regime-switching gate, these ensembles can dynamically shift their predictive weight based on real-time external data, such as extreme weather indicators. By integrating both methods within a weighted framework, hybrid models gain the clarity and stability of econometric tools while benefiting from

the flexibility of machine learning to handle shocks, effectively addressing the limitations of each individual approach [1, 15].

2.4.2 Efficacy of Gradient-Boosted and Deep Learning Ensembles

Empirical studies consistently confirm that hybrid architectures are effective, especially when they combine baseline models with gradient-boosted or deep learning components [1]. In the field of gradient-boosted models, Zhao et al. [26] provided key evidence for this approach with their Temporal Convolutional Network-XGBoost (TCN-XGBoost) hybrid. Their research showed that combining temporal convolutions with XGBoost's ability to model non-linear thresholds resulted in high accuracy for agricultural market data, reaching an RMSE of 0.26 and a MAPE of 5.3%. This performance significantly outperformed standalone ARIMA and SARIMA models, demonstrating that XGBoost is highly effective at capturing the residual shocks that linear models often miss [15].

Similar research has explored deep learning and hybrid models enhanced by decomposition to address extreme market volatility. Pandit et al. [14] used a CEEMDAN-based hybrid architecture and reported a 20% to 62% reduction in forecasting errors compared to standalone ARIMA across various agricultural commodities. Building on this, Zhang et al. [25] and Sun et al. [22] confirmed that sophisticated deep learning ensembles—such as the SVMD-CNN-BiLSTM framework—consistently outperform strictly linear benchmarks by capturing both standard and irregular price movements during periods of high volatility. Additionally, independent evaluations of hybrid models show that while various error metrics are used in comprehensive frameworks [1], focusing on RMSE and MAPE offers a highly reliable assessment of model performance, especially during major price shifts in diverse commodity markets [1, 26].

An objective evaluation of these methodologies reveals certain inherent trade-offs. While hybrid models significantly reduce prediction errors during periods of market shock, they also introduce substantial computational complexity. As Sun et al. [21] note, using sophisticated models with multiple interacting parts increases overall complexity, requiring far more parameter tuning and computational resources than simpler, single-model approaches. Consequently, although the predictive advantages of hybrid models are well-established in the literature, their successful implementation depends on rigorous feature optimization to avoid computational bottlenecks [4, 15].

2.4.3 Adaptation to the Davao City Agricultural Portfolio

This study adopts the core principles established in previous literature, specifically the effectiveness of gradient-boosted residual modeling demonstrated by Zhao et al. [26], and applies them to the vulnerable agricultural market in Davao City. While earlier research confirms the general accuracy of hybrid ensembles [14, 26], this study introduces a localized dimension by implementing a regime-switching mechanism. Since price volatility in the Davao Region is heavily driven by specific, destructive typhoons that disrupt supply chains [17, 19], the static models evaluated in earlier works are structurally inadequate for this context. Consequently, this study aligns with broader literature by using a SARIMAX

baseline for routine market periods. This is specifically chosen over standard SARIMA because the architecture directly incorporates exogenous high-frequency regressors [15], namely daily PAGASA meteorological observations, weekly fuel prices, and the consumer price index, as inputs to the linear component. But this diverges from prior work by employing a dynamic climate-triggered gating mechanism [5]. This gate is explicitly triggered by high-frequency PAGASA climate indicators to activate the XGBoost shock model precisely when non-linear threshold events occur [13, 15].

Moreover, the validation methods used by Manogna et al. [9] and Zhao et al. [26] are closely aligned with the evaluation framework of this study. The Davao vegetable market includes products with widely varying prices, from affordable staples like rice to high-value, volatile crops like cabbage. By adopting the dual-metric approach supported by these researchers, this study ensures the model is strictly penalized for missing extreme price shocks caused by typhoons through the use of Root Mean Squared Error (RMSE) [9]. At the same time, it maintains consistent, scale-independent accuracy across the entire diverse commodity portfolio by using Mean Absolute Percentage Error (MAPE) [9].

Now that the hybrid approach has been shown to align with the research problem, the following section examines the specific computational methods needed to implement this architecture. These include temporal disaggregation, which is used to bridge the data's missing observation problem, and heuristic optimization techniques designed to manage complex climate features.

2.4.4 Computational Mechanics of the Regime-Switching Model

Now that the Hybrid Regime-Switching Approach has been identified as the most suitable framework for modeling climate-driven agricultural price volatility in Davao, this section explores the specific computational mechanics needed to implement it. To address the missing observation problem of data in the local context, multivariate structural time-series disaggregation is first examined as a fundamental data recovery technique [11, 7].

Subsequent sections address the complex challenge of selecting climate features, focusing on the use of Genetic Algorithms for heuristic optimization [3, 4]. Finally, the review synthesizes the mechanics of Extreme Gradient Boosting (XGBoost) for modeling non-linear shocks [2, 26] and the dynamic gating mechanisms necessary for regime identification [5, 15]. Ultimately, this review concludes by establishing the overarching Theoretical Framework, providing the structural foundation for the predictive model presented in Chapter 3.

2.4.5 Temporal Disaggregation

To address the "missing observation problem" discussed in Section 2.2.1, forecasting frameworks must use data recovery techniques that can estimate unobserved high-frequency data from low-frequency official records. The primary method for this process is temporal disaggregation, which was formally established through the multivariate structural time-series frameworks developed by Moauro and Savio [11]. This technique mathemati-

cally breaks down low-frequency series, such as monthly agricultural reports, into higher-frequency intervals like weekly data by using parallel, high-frequency indicator variables. Importantly, the algorithm ensures that the newly estimated weekly data points strictly aggregate or average to match the known monthly baseline, thereby maintaining the statistical integrity of the official records [11].

Recent developments in this field emphasize using multiple indicators to improve the accuracy of data reconstruction [11]. Moauro and Savio [11] demonstrated that including several related data series in a multivariate structural model significantly reduces estimation errors, producing more accurate results than models that rely on a single indicator. Building on this, Mosley et al. [7] introduced regularized methods for high-dimensional data, allowing researchers to incorporate many high-frequency indicators simultaneously without the risk of overfitting. This approach is highly suitable for the local context of this study. By applying multivariate temporal disaggregation using three categories of high-frequency indicator variables which are daily weather observations from PAGASA, weekly fuel prices, and consumer price index, official monthly agricultural price records for the Davao Region can be reliably converted into weekly signals [11, 7]. Fuel prices are included because transport cost fluctuations directly and quickly affect the retail price of crops delivered along Davao’s road networks, particularly when typhoons force rerouting. The consumer price index is included because it tracks the broader inflationary environment that sets the baseline price level within which weather-driven shocks occur, allowing the model to separate structural price trends from sudden climate-driven spikes. For daily weather observations to serve as indicator variables in this process, they must first be converted into the same weekly frequency as the target price data. This is done through temporal aggregation, a straightforward process where daily weather readings such as precipitation, wind speed, temperature are either summed or averaged across each week to produce a weekly weather matrix [11, 7]. This step ensures that both the disaggregated weekly price vector and the aggregated weekly weather matrix share the same time resolution, allowing them to be synchronized into a single high-resolution dataset that the downstream model can use for training and forecasting [15, 12]. This transformation reveals the short-term extreme shocks caused by typhoons, providing the high-resolution target variable (Y) necessary for the shock-modeling components to identify local threshold events.

2.4.6 Heuristic Feature Optimization

A major computational challenge in hybrid agricultural forecasting is the noise and complexity inherent in multi-dimensional feature sets, which require optimization to reduce processing time [4]. When combining data from multiple sources, such as high-frequency climate indicators, macroeconomic variables such as fuel prices and consumer price index, and historical price trends, the resulting feature space becomes highly complex, making it difficult to identify the most relevant variables [15]. Reducing this complexity is essential because unoptimized features increase the computational load and introduce noise, potentially preventing the model from focusing on the key factors that drive market prices and yields [4]. To address this dimensionality issue, modern forecasting frameworks often include metaheuristic optimization layers, most notably Genetic Algorithms (GA),

which use evolutionary search strategies to efficiently identify the most effective subsets of features [4, 3].

The success of Genetic Algorithm (GA)-based feature selection in agricultural research is well-supported by empirical evidence. Chatterjee et al. [4] demonstrated that GA optimization significantly reduces the dimensionality of crop prediction models, creating more efficient feature sets that perform more reliably across different seasonal conditions. Expanding on this, Choudhary et al. [3] found that using GA to tune the hyperparameters of both decomposition and prediction networks overcomes the limitations of manual adjustment. This enables hybrid models to better navigate the challenges of nonstationary and nonlinear price data. Furthermore, in the field of weather forecasting, Shah et al. [20] validated that GA effectively isolates the most relevant non-linear climate interactions, resulting in a notable decrease in forecasting errors compared to models using non-optimized feature sets.

This metaheuristic approach is highly suitable for the architecture of this study. The Davao City vegetable portfolio involves several highly volatile crops that interact with varying climate thresholds, creating an expansive search space. By implementing a GA-based optimization layer before the final model training, the system can efficiently navigate this search space to identify the most effective predictors and feature subsets [4]. This ensures that only the most significant indicators are passed to the XGBoost component, effectively filtering out noise and irrelevant data to prevent the model from overfitting [4].

2.4.7 Extreme Gradient Boosting (XGBoost) for Shock Modeling

Once the climate features have been optimized, the hybrid framework requires a powerful computational engine to link these complex inputs to non-linear price movements. Research identifies Extreme Gradient Boosting (XGBoost), introduced by Chen and Guestrin [2], as a highly effective tool for this task, particularly for identifying subtle, non-linear relationships between diverse weather and market factors [15]. Unlike some deep learning models that can be difficult to interpret, XGBoost is a sophisticated ensemble of decision trees that uses regularized gradient boosting [15, 2]. This regularization helps prevent the model from becoming overly complex, offering essential protection against overfitting when working with noisy or volatile data [2]. Additionally, when used with Explainable AI tools like SHAP (SHapley Additive exPlanations), researchers can identify and measure exactly which variables are influencing the model's price forecasts [15].

In agricultural and economic forecasting, XGBoost has become the leading framework for analyzing structured tabular data [15, 2]. Zhao et al. [26] and Mulla and Usha [12] confirm that when XGBoost is trained on high-frequency datasets, it identifies non-linear threshold effects with much greater precision than traditional models using low-frequency monthly averages. By using data at a weekly resolution, enabled by the temporal disaggregation described in Section 2.4.5, these gradient-boosted trees can effectively learn the specific weather-related tipping points that lead to sudden spikes in retail prices.

These methodological strengths make XGBoost an ideal choice for modeling the Davao City vegetable portfolio. Localized price volatility in the region is driven by several interacting variables, including extreme precipitation, macroeconomic fuel prices, and historical price patterns. Because XGBoost can model these complex interactions without being limited by rigid linear assumptions, it functions effectively as the dedicated "shock modeling" component within the proposed ensemble [15]. Additionally, when integrated with Explainable AI frameworks like SHAP, the model translates complex algorithmic findings into practical policy insights. This allows local stakeholders to identify precisely which climate drivers are most responsible for destabilizing the Davao market [15].

2.4.8 Regime Identification and Ensemble Weighting

While hybrid architectures possess both linear and non-linear forecasting capabilities, their effective deployment depends on a mathematical mechanism to determine which component should drive the forecast at any given time. The theoretical foundation for dynamic deployment in commodity pricing was established by Yuan and Kwon [24], who showed that Markov regime-switching models effectively capture structural shifts between distinct market states. Building on this for agricultural applications, Montaña et al. [5] demonstrated that regime-switching models can actively recognize transitions between low-volatility and high-volatility production phases. This approach is especially relevant in the Philippine context; Montaña et al. [5] provided empirical evidence that structural volatility shifts in local agricultural production are specifically triggered by climate events like El Niño, La Niña, and typhoons. Since static linear models consistently fail to detect these specific regime changes, a dynamic transition mechanism is necessary [5].

To put this transition into practice, modern frameworks use a continuous gating mechanism driven by high-frequency external indicators [5]. As Patro et al. [15] validated in their transparent forecasting framework, using multi-factor driver analysis to explicitly model regime persistence results in predictions that are both more accurate and easier for policymakers to interpret. In practice, this gating logic monitors local climate indicators—such as PAGASA precipitation thresholds—alongside macroeconomic factors. When these variables stay within historical limits, the system operates in a 'Normal Regime,' giving the primary predictive weight to the SARIMAX baseline to capture routine seasonal trends and the gradual inflationary environment independently of climate shocks [5].

The moment a climate anomaly exceeds a predefined critical threshold, an indicator function (I_{shock}) activates, transitioning the architecture into the "Shock Regime." In this state, the ensemble's weighting dynamically shifts to favor the XGBoost component [15]. This ensures that the non-linear learner drives the forecast exactly when abrupt, typhoon-driven price spikes are mathematically imminent [15, 26]. By employing this environment-responsive weighting strategy, the hybrid model maintains high accuracy during both stable periods and severe disruptions in the Davao City vegetable portfolio.

2.5 Summary and Research Gap

Having established the theoretical problem, evaluated the prevailing approaches, and synthesized the required computational methodologies, this final section summarizes the current Body of Knowledge (BoK). By highlighting key research studies, it explicitly identifies the remaining gaps in the literature, particularly the need for high-frequency, climate-integrated models in regional Philippine contexts, and justifies the present study's localized application to the Davao City agricultural market. This synthesis ensures that the proposed hybrid architecture is both theoretically grounded and specifically tailored to address the unique volatility patterns found in the Davao region.

2.5.1 Summary of Key Research Studies

Table 1: Summary of Methodologies and Key Findings

Methodology / Model	Application Context & Key Findings
Classical Econometric Models (ARIMA, SARIMAX)	Found fundamentally inadequate for capturing the sudden volatility and nonlinear dynamics inherent in agricultural markets [12, 21].
Learning-Based Hybrid Approaches (TCN+XGBoost, SARIMAX+XGBoost)	Validated superiority in reducing forecasting errors during high volatility [14]. Pairing TCN with XGBoost captures nonlinear threshold effects [26]. Integrating a linear baseline (SARIMAX) with XGBoost creates a robust ensemble for both stable and disrupted market regimes [15].
Temporal Disaggregation	Established the mathematical framework proving high-frequency indicator data which includes weather observations, fuel prices, and consumer price index can successfully estimate unobserved high-frequency data points from low-frequency aggregates [11, 7].
Metaheuristic Optimization (Genetic Algorithms)	Effectively isolates optimal predictive features and filters irrelevant data from multi-dimensional datasets [4]. Critical for optimizing the complex hyperparameters of decomposition and deep learning components [3].
Regime-Switching Architectures	Established the theoretical and empirical necessity of dynamic gating; demonstrates that traditional static models fail to capture structural volatility breaks and climate-driven regime shifts in agricultural production [5, 24].

The reviewed literature highlights a clear evolutionary trajectory in agricultural price forecasting, beginning with the identification of the structural limitations of classical models. Foundational studies by Sun et al. [22] and Mulla and Usha [12] established that purely linear econometric models, such as ARIMA and SARIMA, are fundamentally inadequate for capturing the sudden volatility and nonlinear dynamics inherent in agricultural markets. To address these gaps, Zhao et al. [26] and Pandit et al. [14] validated the

superiority of learning-based hybrid approaches, demonstrating that combining advanced neural networks with secondary algorithms significantly reduces forecasting error during periods of high volatility. Specifically, Zhao et al. [26] showed the effectiveness of pairing temporal convolutional networks with Extreme Gradient Boosting (XGBoost) to capture nonlinear threshold effects. Building on this, Patro et al. [15] established that integrating a linear baseline like SARIMAX with the non-linear capabilities of XGBoost creates a robust ensemble capable of effectively managing both stable and disrupted market regimes.

The literature highlights several critical methodologies for operationalizing these hybrid models. Moauro and Savio [11] alongside Mosley et al. [7], established the mathematical framework for temporal disaggregation, proving that high-frequency indicator data can successfully estimate unobserved high-frequency data points from low-frequency aggregates. Concurrently, the necessity of metaheuristic optimization was validated by Chatterjee et al. [4], who demonstrated that Genetic Algorithms (GA) effectively isolate optimal predictive features and filter irrelevant data from multi-dimensional datasets. Expanding on these applications, Choudhary et al. [3] showed that GA is also critical for optimizing the complex hyperparameters of decomposition and deep learning components. Finally, Yuan and Kwon [24] and Montañó et al. [5] established the theoretical and empirical necessity of regime-switching architectures, demonstrating that traditional models fail to capture structural volatility breaks and climate-driven regime shifts in agricultural production. Together, these established methodologies provide the theoretical foundation for implementing a dynamic, climate-triggered gating mechanism to accurately forecast market fractures.

2.5.2 Research Gap and Justification

Despite these methodological advancements, a critical synthesis of the existing literature reveals a distinct gap regarding the localized application of high-frequency climate anomalies to agricultural retail price forecasting. Specifically, several deficiencies remain that this study directly addresses:

- **Geographic and Infrastructural Specificity:** Existing frameworks, such as those established by Li et al. [6], primarily focus on macro-level economic outcomes but inherently overlook the unique structural vulnerabilities present in the Philippines. These generalized models fail to account for the severe disruption of transportation networks, blocked roads, and localized logistical bottlenecks that routinely trigger sudden agricultural price shocks in vulnerable areas such as the Davao Region [17, 16]
- **The Granularity and Shock Modeling Deficit:** Although traditional econometric models systematically fail to capture extreme, non-linear market volatility [12, 22], existing temporal disaggregation frameworks [11, 7] have yet to be applied to identify typhoon-induced agricultural price shocks specifically within the Davao market.
- **Localized Regime Modeling:** While regime shifts in the broader context of Philippine agricultural production have been established [5], there is a lack of research integrating these insights into a high-frequency, weather-price hybrid model using XGBoost specifically for Davao City.

This research justifies the implementation of a Regime-Switching Ensemble Architecture as a necessary contribution to the Body of Knowledge. By integrating temporal disaggregation [11, 7], heuristic feature optimization [3, ?], and a dynamic climate gate [15], the model overcomes the excessive smoothing found in traditional techniques [23, 10]. By filling these gaps, the study moves beyond simple prediction to provide a specialized mapping function $f : X \rightarrow Y$ evaluated using dual metrics [26, 9]. This function explicitly reflects the true extent of price volatility in a climate-sensitive tropical market, providing actionable, data-driven insights for farmers, logistics planners, and policymakers in the Davao Region [26, 15].

2.6 Theoretical Framework

The predictive architecture of this study is anchored in a synthesis of econometric time-series theory, evolutionary computation, and machine learning ensemble dynamics. Rather than relying on a single algorithmic paradigm, this framework integrates distinct theoretical models to address the interconnected challenges of data frequency, high-dimensional feature spaces, and non-linear market shocks. The overarching theoretical foundation is built upon the following five interconnected models.

2.6.1 Theory of Temporal Disaggregation

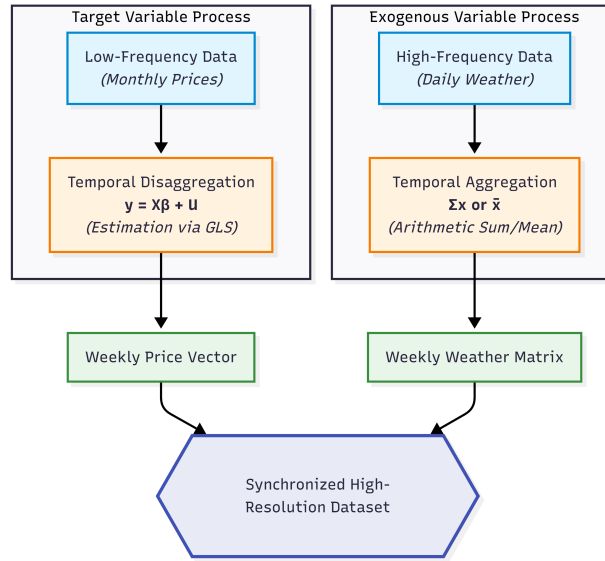


Figure 1: Theory of Temporal Disaggregation

As shown in the diagram, temporal disaggregation utilizes two parallel tracks to construct a synchronized high-resolution dataset [11, 7]. The first track, the Target Variable Process, converts low-frequency monthly price information into a weekly price vector by employing the regression model $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}$, estimated through Generalized Least Squares (GLS). Concurrently, the Exogenous Variable Process manages high-frequency

daily weather data by applying temporal aggregation, using either arithmetic summation ($\sum x$) or averaging ($\bar{x}$), to align the information into a weekly weather matrix. These distinct outputs are ultimately integrated into a single, synchronized high-resolution dataset to facilitate downstream modeling [7, 11].

This dual-track architecture is rooted in the methodologies developed by Moauro and Savio [11] and Mosley et al. [7]. Specifically, Moauro and Savio [11] utilized multivariate structural time series models to estimate high-frequency economic series from aggregates across 12 OECD countries, ensuring that disaggregated estimates remained consistent with established low-frequency benchmarks. Mosley et al. [7] advanced this concept to address high-dimensional data, such as CO_2 emissions, by integrating the Chow-Lin regression framework with LASSO regularization to identify significant high-frequency indicators. This study adopts these techniques to reconstruct weekly vegetable prices for Davao City from official monthly data, leveraging three categories of high-frequency exogenous indicators: daily weather observations from PAGASA, weekly fuel prices, and the consumer price index [7, 11].

2.6.2 Evolutionary Theory of Heuristic Feature Optimization

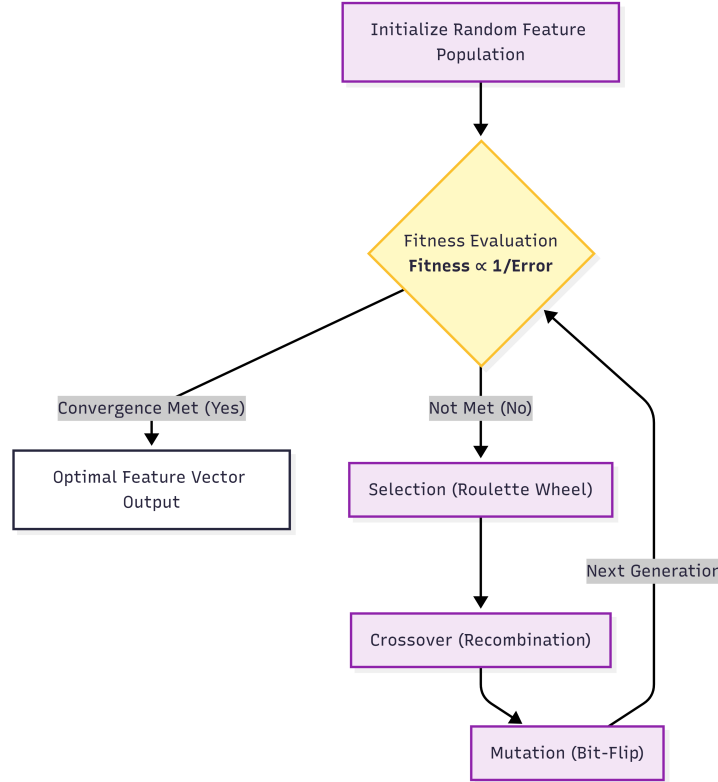


Figure 2: Evolutionary Theory of Heuristic Feature Optimization

As illustrated in the diagram above, the Genetic Algorithm (GA) begins by initializing a random population of candidate feature subsets, each evaluated by a fitness function defined as $\text{Fitness} \propto 1/\text{Error}$, meaning subsets that minimize prediction error are assigned higher fitness scores [4]. If convergence is not yet met, the algorithm proceeds

through three evolutionary operators: tournament selection, which retains the highest-performing subsets; crossover (recombination), which combines features from two parent subsets to produce offspring; and mutation (random swap), which introduces diversity by randomly altering individual features. This cycle repeats across successive generations until the convergence criterion is satisfied, at which point the optimal feature vector is output for downstream modeling [4].

This methodology is rooted in the research of Chatterjee et al. [4], who applied an Enhanced Genetic Algorithm (EGA) to forecast crop yields using multi-dimensional environmental datasets, including variables such as temperature, humidity, soil pH, and nutrient concentrations. Chatterjee et al. [4] demonstrated that by integrating tournament selection and elite preservation, the EGA successfully avoids premature convergence—a common pitfall in standard GA implementations. Their findings showed that this approach identifies more relevant feature subsets, which increased gradient boosting model accuracy from 71.871% to 93.009% and decreased mean squared error from 0.962 to 0.534. The proposed study adopts this same evolutionary framework to analyze the high-dimensional climate and macroeconomic feature space from daily PAGASA observations, ensuring that the XGBoost shock-modeling component receives only the most statistically significant predictors [4].

2.6.3 Theory of Regime-Switching Dynamics and Market States

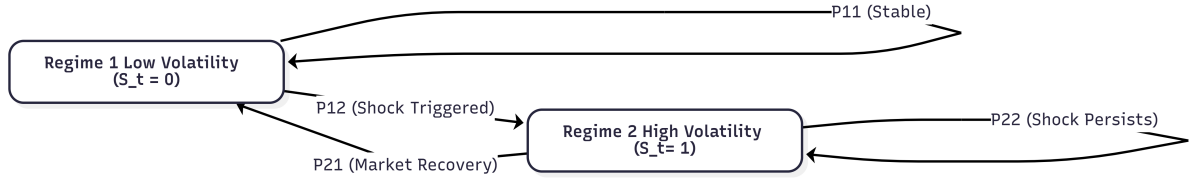


Figure 3: Theory of Regime-Switching Dynamics and Market States

As depicted in the diagram, the regime-switching framework characterizes the agricultural market through two distinct states: Regime 1 (Low Volatility, $S_t = 1$) and Regime 2 (High Volatility, $S_t = 2$), which are defined by four transition probabilities [5, 24]. P_{11} signifies the likelihood of maintaining stability within the Normal Regime, while P_{12} indicates a shock-driven transition into the High Volatility Regime. In contrast, P_{22} accounts for the persistence of a shock within the High Volatility Regime, and P_{21} represents the market's recovery as it transitions back to the Normal Regime. These probabilities collectively form a first-order Markov chain, where the state of the current regime is dependent only on the state immediately preceding it [5].

The theoretical foundation of this structure is came from the research of Montaña et al. [5] and Yuan and Kwon [24]. In their analysis of Philippine corn production from 2010 to 2025, Montaña et al. [5] utilized a two-state Hidden Markov Model on quarterly production data. Their findings revealed that the system exhibited near-deterministic transitions between low- and high-volatility regimes, primarily driven by climatic events such as El Niño, La Niña, and typhoons. The 2-state HMM demonstrated superior performance over the 1-state specification, as evidenced by a lower AIC (1,575.499 versus

1,696.925), thereby confirming the empirical existence of discrete regime structures in Philippine agricultural data. Furthermore, Yuan and Kwon [24] established that these structural market shifts can be formally captured through Markov switching influenced by exogenous environmental variables. This study applies that framework to Davao vegetable prices, employing PAGASA climate thresholds as the exogenous triggers for transitions between the Normal and Shock Regimes [5, 24].

2.6.4 Theoretical Mechanics of Dual-Regime Hybrid Ensembles

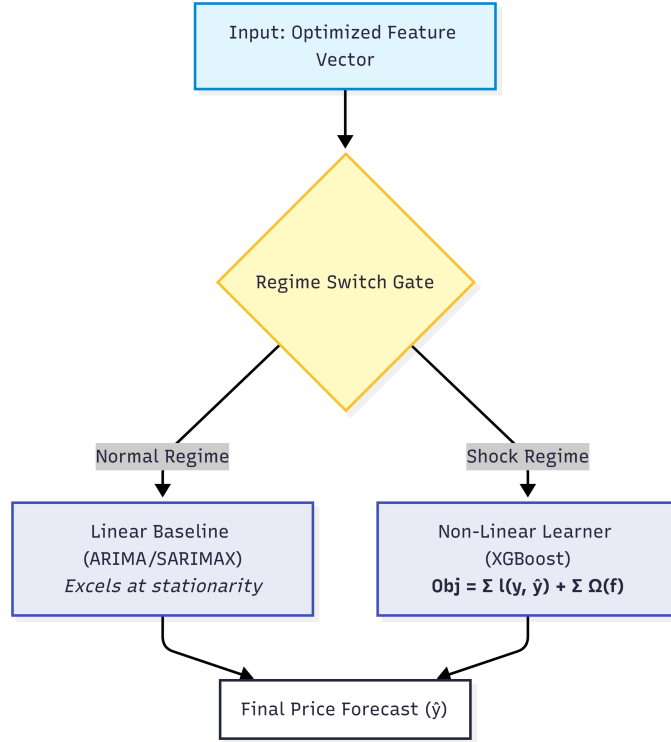


Figure 4: Theoretical Mechanics of Dual-Regime Hybrid Ensembles

As shown in diagram 4, the predictive mechanism employs a dual-engine architecture to address the two-fold volatility of agricultural markets. Regarding crop price forecasting, Mulla and Usha [12] established that classical linear baselines like ARIMA or SARIMAX are theoretically optimal for extracting routine, stationary seasonal trends but lack the mathematical capacity to project sudden structural breaks. To address this limitation within highly volatile agricultural contexts, Zhao et al. [26] utilized Extreme Gradient Boosting (XGBoost) as a non-linear learner. By applying regularized gradient descent over an ensemble of decision trees, Zhao et al. [26] demonstrated how XGBoost effectively models complex, non-linear threshold effects without imposing parametric assumptions—proving its capacity to maintain predictive accuracy during severe market disruptions.

Instead of deploying these models independently, the architecture utilizes an en-

semble combination layer formalized by Patro et al. [15] within their multi-factor crop price forecasting framework. This method dynamically integrates linear and non-linear outputs via the ensemble weighting equation $\hat{X} = \partial_1 g_{stat} + \partial_2 g_{ml}$, where ∂_1 and ∂_2 represent modulating coefficients. In Patro et al.’s [15] formulation, the statistical component g_{stat} is instantiated as SARIMAX rather than SARIMA, specifically because the exogenous regressor interface of SARIMAX allows the baseline model to directly ingest the same high-frequency weather and macroeconomic indicators used in the temporal disaggregation stage — an architectural requirement that standard SARIMA cannot fulfill [15]. As managed by the Regime Identification Gate, these coefficients are actively adjusted: during normal weather, ∂_1 is prioritized to maintain the stability of the baseline g_{stat} , whereas a climate threshold breach amplifies ∂_2 to shift the predictive weight toward the XGBoost component g_{ml} [15, 26]. This dynamic weighting ensures the final price forecast leverages the specific mathematical strengths of both paradigms while effectively mitigating their individual vulnerabilities.

2.6.5 Theoretical Basis for Dual-Metric Evaluation

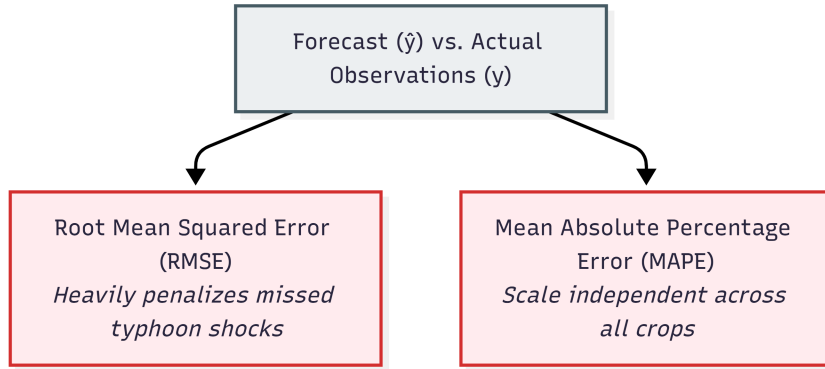


Figure 5: Theoretical Basis for Dual-Metric Evaluation

As illustrated in Figure 5, evaluating a dynamic regime-switching architecture requires a theoretically robust approach to error measurement, as relying on a single metric introduces systematic bias across stable and shock-driven market conditions. In the context of deep learning for agricultural commodity price forecasting, Manogna et al. [9] established the necessity of a dual-metric strategy to accurately assess predictive models across heterogeneous crop values, such as onions and tomatoes. To mathematically capture and heavily penalize extreme price shocks such as those caused by typhoons, Manogna et al. [9] utilized Root Mean Squared Error, formulated as

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{Y}_i - Y_i)^2}.$$

By squaring the magnitude of prediction errors before averaging them, RMSE serves as a rigorous penalty function. As indicated in the "Shock Penalty" node of the diagram, this specific mathematical property evaluates the non-linear learner’s capacity

to precisely map highly volatile market disruptions by ensuring that significant price deviations are not overlooked.

To complement this sensitivity to shocks, Manogna et al. [9] concurrently utilized Mean Absolute Percentage Error, formulated as:

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{Y_i - \hat{Y}_i}{Y_i} \right| \times 100,$$

to provide a scale-independent assessment of baseline accuracy. Because MAPE expresses errors as a relative percentage of actual values rather than absolute currency units, it allowed Manogna et al. [9] to consistently evaluate and compare forecasting performance across commodities with differing price magnitudes. As depicted in the "Scale Independence" branch of the diagram, this approach is vital for the proposed framework's application to Davao City's diverse vegetable portfolio, ensuring that both high-value crops and affordable staples are evaluated fairly. Together, these two metrics theoretically validate the architecture's dual requirements: RMSE ensures the model is strictly penalized for failing to predict typhoon-driven price spikes, while MAPE guarantees consistent accuracy during routine seasonal regimes.

Chapter 3

Methodology

3.1 Conceptual Framework

The study’s conceptual framework outlines the data transformation, feature optimization, and hybrid predictive architecture used to forecast agricultural commodity prices. It specifically demonstrates how various data sources, including historical prices, weather conditions, and macroeconomic indicators, are mathematically synchronized and processed to isolate the impact of extreme climate anomalies on the local vegetable portfolio.

3.1.1 Phase 1: Data Resolution

The data resolution phase involves preprocessing various system inputs to ensure they are properly aligned. Addressing the ”missing observation problem” found in official monthly reports is a mathematical necessity; simply reducing all variables to a low frequency would erase the short-term shocks the system is built to forecast [11]. In this research, low-frequency endogenous data, specifically Davao City’s monthly retail prices, undergoes Multivariate Temporal Disaggregation via Generalized Least Squares (GLS) Regression to estimate unobserved weekly variations. In contrast, high-frequency exogenous data, such as daily weather indicators from PAGASA, is processed through Temporal Aggregation using arithmetic sums or averages to condense the information into weekly intervals. Finally, macroeconomic variables like weekly fuel prices and the consumer price index are integrated directly into the synchronized dataset as weekly control variables. Unlike the monthly price data that requires disaggregation or the daily weather data that requires aggregation, these macroeconomic indicators are sourced or interpolated at a weekly resolution. This allows them to be fed directly into the High-Resolution Weekly Dataset without needing a frequency conversion step.

3.1.2 Phase 2: Dimensionality Reduction

The dimensionality reduction process uses a standard Genetic Algorithm (GA) to perform heuristic feature optimization. Managing large, multi-dimensional climate datasets can

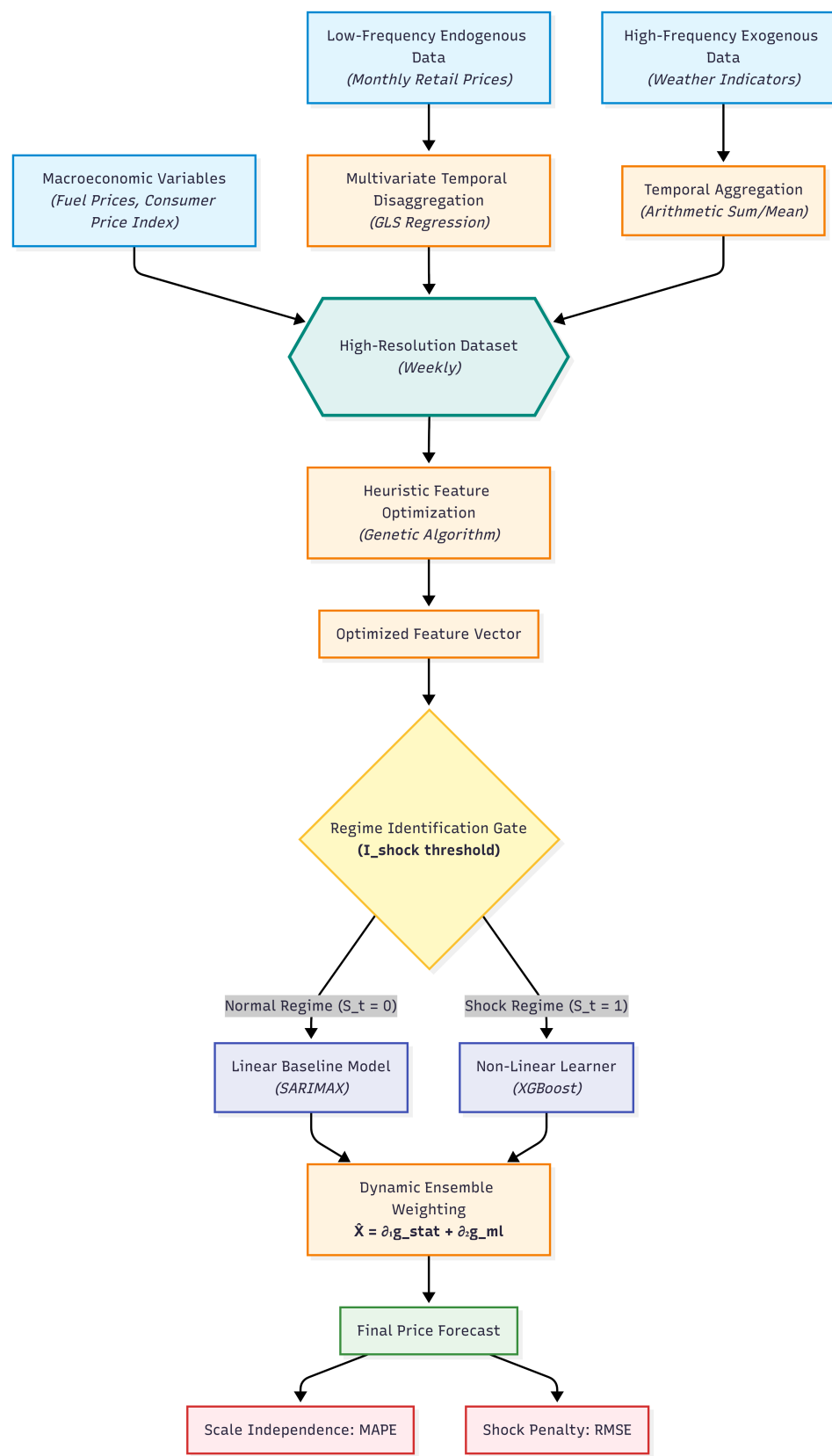


Figure 6: Conceptual Framework of the Dual-Regime Predictive Engine

lead to spectral noise and an unmanageable number of potential feature combinations. This requires a metaheuristic approach to find the global optimum efficiently without the need to test every single permutation [4]. Within this pipeline, the GA examines the high-resolution weekly dataset by using an evolutionary cycle of selection, crossover, and mutation to remove irrelevant weather noise. This method effectively isolates the most statistically significant predictive climate-price lags, which then forms the optimized feature vector.

3.1.3 Phase 3: Regime-Switching Logic

The regime-switching logic acts as a dual-engine predictive core controlled by an algorithmic gating mechanism. Using only one computational approach is mathematically insufficient to handle the two-fold nature of agricultural volatility. Traditional models often fail to predict sudden structural breaks [12], while purely non-linear models risk overfitting during stable, routine seasons [26]. This framework employs a Regime Identification Gate that is activated by a specific climate threshold (I_{shock}). When weather conditions remain normal ($S_t = 0$), the optimized feature vector is directed to the linear baseline model, which is SARIMAX. If an extreme threshold is crossed ($S_t = 1$), the vector is instead sent to the non-linear learner, XGBoost. The results from both engines then merge at a Dynamic Ensemble Weighting node, where the equation $\hat{X} = \partial_1 g_{stat} + \partial_2 g_{ml}$ mathematically adjusts the relative influence of each model based on the current environmental regime [15]. The climate thresholds θ_P and θ_W are operationally defined using PAGASA’s Tropical Cyclone Warning Signal (TCWS) classifications. Specifically, the model utilizes Signal No. 1 sustained wind speed thresholds (≥ 30 km/h) and their corresponding precipitation accumulation benchmarks, following the methodology established in prior Philippine agricultural climate studies [5, 13].

3.1.4 Phase 4: Output and Validation

The output and validation phase produces the final price forecast and assesses its mathematical accuracy through a dual-metric strategy. Using a single error metric can lead to systematic bias when evaluating hybrid models across both stable and volatile market conditions [9]. To address this, the final price forecast is measured simultaneously using Mean Absolute Percentage Error (MAPE) and Root Mean Squared Error (RMSE). MAPE provides scale independence [9], which ensures a fair accuracy assessment for both high-value and affordable crops in the Davao portfolio. At the same time, RMSE serves as a strict shock penalty that heavily penalizes the architecture for any failure to capture significant [9], typhoon-induced price spikes.

To ensure the evaluation metrics reflect genuine predictive ability rather than just fitting the training data, model validation follows a walk-forward expanding-window protocol. The full nine-year dataset is split so that the first 80 percent of weekly observations form the training set, while the remaining 20 percent is reserved as a held-out test set. Instead of using a single static split, the walk-forward approach gradually increases the training window by one week at each step. It generates a new one-step-ahead forecast before the window moves forward. This protocol strictly maintains chronological order,

which ensures the model cannot access future price information during prediction and prevents data leakage across the training and testing boundary [11, 15]. Both RMSE and MAPE are calculated for all out-of-sample forecast steps and then averaged to produce the final performance metrics. This ensures the evaluation accurately reflects the model's behavior during both stable seasonal periods and typhoon-driven price shocks within the test set.

This multi-stage conceptual pipeline directly addresses the structural limitations of local price forecasting. By mathematically synchronizing different data frequencies, isolating the most effective features, and dynamically adjusting the computational focus during extreme weather events, this architecture provides a highly adaptive solution to agricultural market volatility. These elements work together to establish a solid foundation for the formal research design.

3.2 Research Design

This study utilizes a quantitative and predictive computational research design, featuring a hybrid model that integrates machine learning with econometric forecasting. The choice of a dual-engine architecture is necessary because climate-sensitive agricultural markets experience two distinct types of volatility. Traditional time-series models are effective for capturing regular seasonal patterns, but they lack the mathematical capacity to predict sudden and sharp market disruptions [12]. Conversely, relying solely on non-linear machine learning models can lead to significant overfitting during periods of market stability [26]. Within the context of the Davao City agricultural market, this design creates a robust prediction system by shifting between linear and non-linear logic based on environmental triggers.

3.3 Data Collection

Endogenous Target Data Historical monthly retail prices for vegetables serve as the main target variable for this system. It is necessary to obtain this baseline data before using temporal disaggregation algorithms to ensure that the estimated high-frequency variables do not conflict with established low-frequency constraints [11]. This information will be sourced from the Philippine Statistics Authority (PSA) OpenSTAT database, covering the specific vegetables and timeframe selected for the study.

Exogenous High-Frequency Data Daily weather records act as the high-frequency environmental triggers for this study. Detailed climate data is theoretically required to establish the initial state transition probabilities in a regime-switching architecture [5]. This information, which includes total daily rainfall, maximum sustained wind speeds, and mean weekly temperature, will be sourced directly from regional Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAGASA) stations located in the Davao Region.

Macroeconomic Control Data The monthly Consumer Price Index (CPI) and regional retail fuel prices serve as system controls. Integrating these variables is necessary to distinguish climate-related supply chain volatility from broader national inflation trends, which prevents the model from incorrectly attributing the cause of price shifts. These datasets will be sourced from the PSA and Department of Energy (DOE) web portals and merged directly into the synchronized dataset.

The data collection process adheres to a rigorous technical workflow to guarantee complete data integrity before formal preprocessing begins.

1. **Extraction:** Historical data covering the designated nine-year period (2018 - 2026) will be retrieved in CSV and Excel formats from the official government web portals of the PSA, PAGASA, and DOE.
2. **Alignment:** The various datasets will be algorithmically audited for temporal alignment. This involves strictly verifying that the start and end dates of the weather and macroeconomic data perfectly match the timeframe of the agricultural price reports.
3. **Imputation:** Missing data points in the continuous time series, such as unrecorded daily temperatures caused by local station sensor failures, will be handled using k-Nearest Neighbors (k-NN) imputation or linear interpolation. This ensures the data is complete and consistent before starting the temporal aggregation and disaggregation processes.

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